

Preferential conduction in the atria: clinical implications

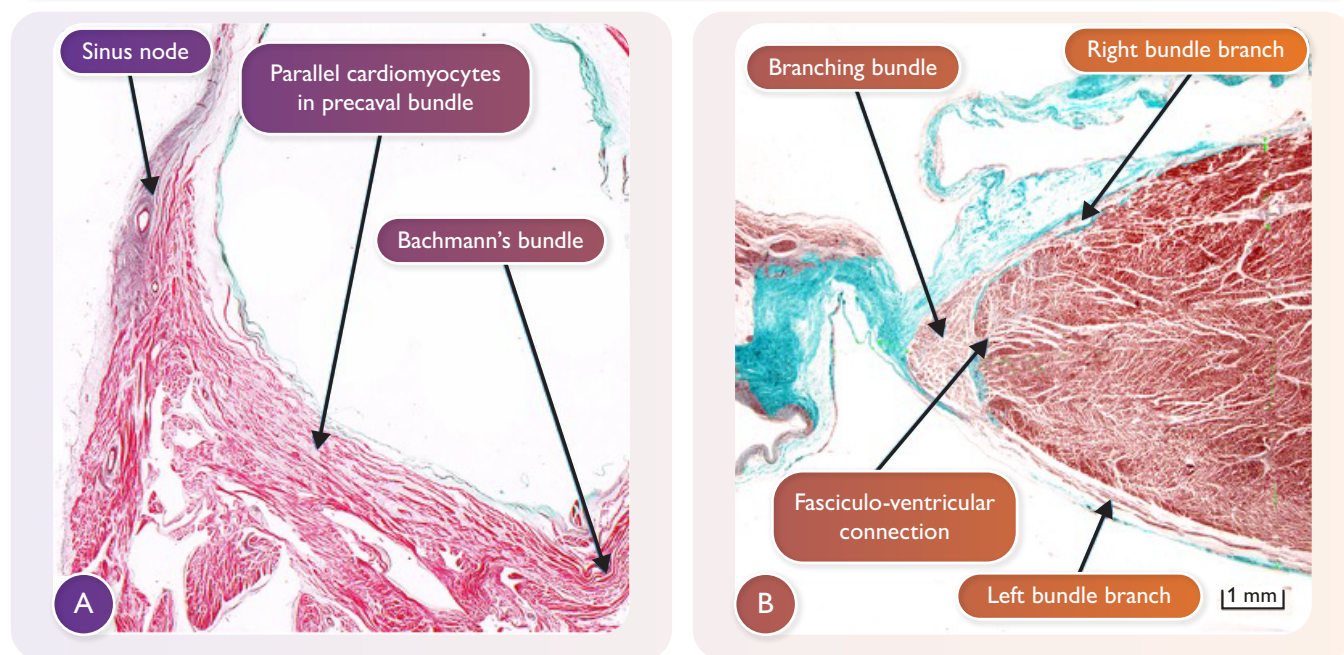
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Graphical Abstract

Preferential vs specialized conduction in the heart



As is shown in panel A, the conduction of the impulse generated by the sinus node through the atrial walls is dependent on the parallel alignment of the aggregated working atrial cardiomyocytes. This is in marked contrast to the feature that permits recognition of the cardiomyocytes underscoring passage of the impulse so as to activate the ventricular myocardium. As shown in panel B, this depends on the presence of ventricular cardiomyocytes, themselves histologically distinct from the adjacent working cardiomyocytes of the ventricular septum, but additionally insulated from the ventricular myocardium by the presence of fibrous sheaths. By virtue of the presence of the sheaths, the ventricular myocardium is activated from apex to base. The image in panel B, however, also illustrates a fasciculo-ventricular connection, which provides a direct connection to the septal crest

mm, millimeter

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Abstract

There are areas in the heart, and especially the atrial walls, where cardiomyocytes are aggregated together in a parallel fashion, producing preferential pathways for conduction. Although aggregated together, however, the cardiomyocytes lack the insulation that is the characteristic feature of the 'specialized' ventricular conduction system. This review presents the anatomical and histological evidence that emphasizes the fundamental differences between preferential conduction across the atrial walls, as opposed to 'specialized' activation of the ventricular myocardium. With the advent of novel techniques, it may well prove possible to identify particular features of the histologically indistinct cardiomyocytes making up the atrial walls that influence their speed of conduction and hence perhaps underscore a mode of 'functional' specialization, but as yet such evidence remains lacking.

Keywords Preferential conduction • Functional specialization • Insulated conducting tracts • Ventricular conduction system

Introduction

All cardiomyocytes making up the walls of the cardiac chambers have the capacity to conduct. If diseased or abnormal, they also have the capacity to give rise to, or become part of a circuit for, cardiac arrhythmias. A specific subset of cardiomyocytes, however, is currently recognized as providing the specialized 'conduction system' of the heart.¹ The insulated 'tracts' within this system are found only within the ventricular mass.² Shortly after the discovery of the conduction system itself, however, suggestions were made that similar tracts might exist within the atrial walls. The problem in adjudicating this suggestion was that, when assessed grossly, there was no way of distinguishing the alleged 'specialized' cells from the so-called working cardiomyocytes, with the latter cells making up the greater part of the chamber walls. Thus, a series of criteria was proposed to permit recognition, when using the light microscope, of the characteristics of the cardiomyocytes making up the system specialized to initiate and coordinate cardiac contraction.^{3,4} Now known as the Aschoff-Monckeberg criteria,⁵ the first of these rules dictated that the alleged 'specialized' cardiomyocytes should be histologically discrete when assessed relative to adjacent working cardiomyocytes (*Figure 1, panels A–D*). Those investigating the nature of the cardiomyocytes prepared their datasets by cutting multiple sections in serial fashion. The second proposed criterion, therefore, was the ability to follow the 'specialized' cardiomyocytes from section to section in the serially sectioned datasets. The final criterion, which is now recognized to be specific for the definition of tracts, as opposed to the cells aggregated into nodes, was that the cardiomyocytes considered to be making up a tract should be insulated by fibrous sheaths from their working neighbours (*Figure 1, panels C & D*). The exemplars for insulated tracts chosen by those initially producing the criteria were the ventricular components of the atrioventricular conduction axis as had initially been described by Tawara.⁶ When applying these criteria for examination of histological sections, the cardiomyocytes currently accepted as being 'specialized', and hence representing the conduction system of the heart are, indeed, discrete. They can be followed from section to section when assessed in the histological datasets prepared in serial fashion. And, when considering the ventricular components of the conduction axis, the specialized cardiomyocytes are insulated by fibrous sheaths from their working

neighbours (*Figure 1, panels C & D*).⁵ The 'specialized' cardiomyocytes recognized histologically as generating and disseminating the cardiac impulses in the normal heart, therefore, can be considered in terms of the sinus node (*Figure 1A*) and the atrioventricular conduction axis (*Figure 1, panels B–D*).^{5,6} In the heart, however, there are other areas where cardiomyocytes can be seen to be aggregated together in a parallel fashion. They are known to produce preferential pathways for conduction. The cardiomyocytes themselves, however, lack any features which, when assessed using standard histological techniques, permit them to be described as anything but ordinary working cardiomyocytes.

Sinus to atrioventricular nodal 'pathways'

Perhaps the most notable example of the areas providing 'preferential conduction' is that taking place within the atrial walls between the sinus and atrioventricular nodes. The sinus node has irregular margins. Although supported within an obvious fibrous matrix, it is not insulated at its margins by a sheath of fibrous tissue (*Figure 1A*).^{7,8} Instead, its histologically specialized cardiomyocytes merge at the margins of the node with the muscular sleeve of the superior caval vein and with the working myocardium of the terminal crest. Cardiac electrical activity starts with the spontaneous excitation of 'pacemaker' cells in the sinus node. Pacemaker automaticity is due to spontaneous diastolic repolarization (Phase 4), which generates rhythmic action potentials. These potentials then determine the heart rate through various currents, including the I_f current. The process of excitation then depolarizes the working cardiomyocytes, providing a wave within the atrial walls which moves towards the atrioventricular node via pathways made up of well-aligned working atrial cells.^{7,9} We make no claims for originality in emphasizing the role of working cardiomyocytes aggregated together in parallel fashion as providing the preferential pathways. In fact, the evidence of the ordinary nature of the myocardium making up the atrial walls was provided over a century ago and emphasized by Bachmann when describing his bundle.⁸ Despite the existence of this evidence, however, the suggestion was subsequently made that some of the cardiomyocytes providing these pathways were anatomically specialized,

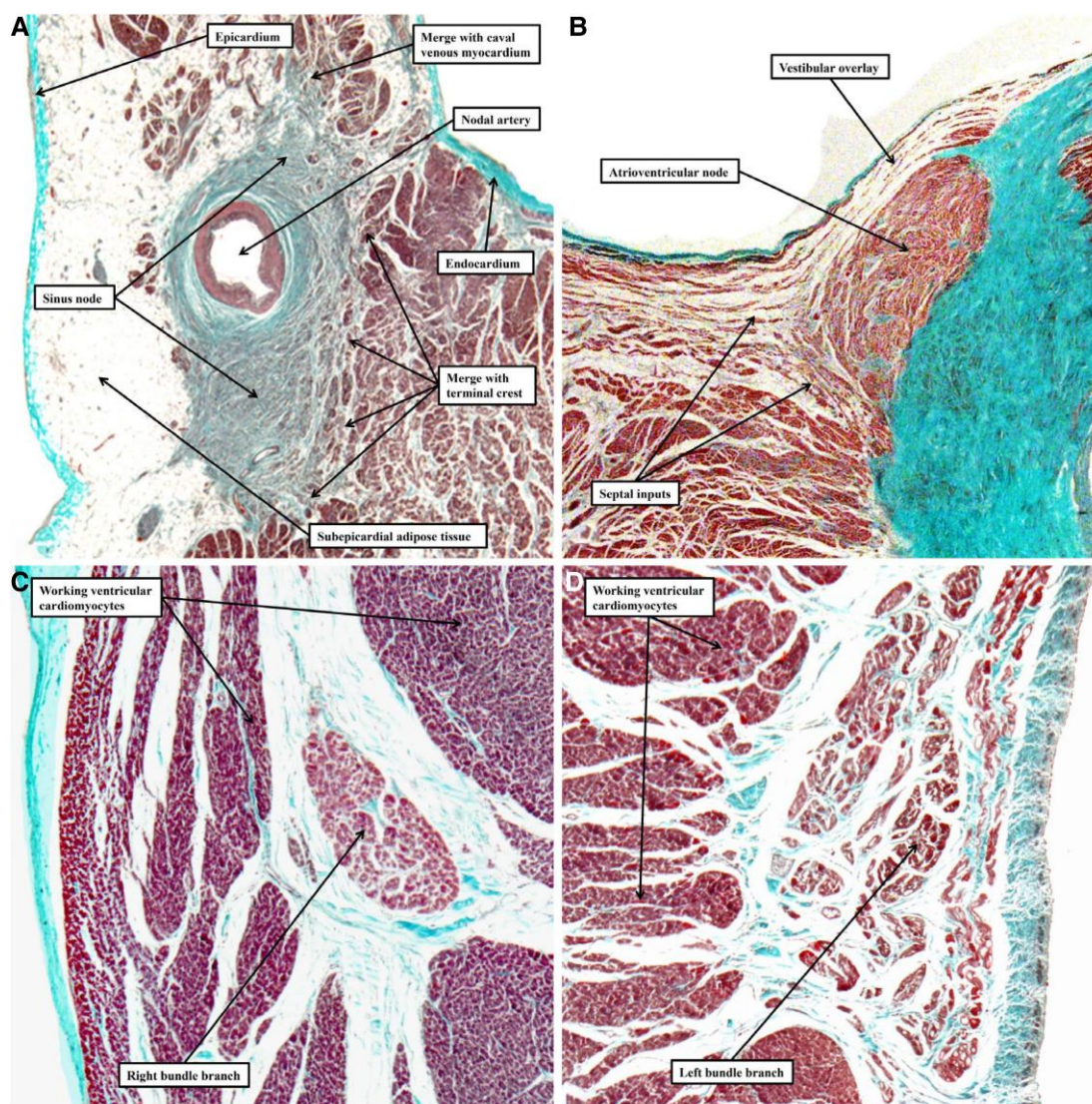


Figure 1 All the sections in the panels are stained using the trichrome technique. This stain colours fibrous tissue in blue-green and myocardium in red-brown. The panels show how it is possible to distinguish the cardiomyocytes making up the sinus and atrioventricular nodes on the basis of the first criterion proposed by Aschoff and Monckeberg for histological specialization, namely, the distinction between the histological appearance of the 'specialized' cardiomyocytes and their adjacent 'working' neighbours. The second criterion is the ability to follow the identified structure in sections cut in serial fashion. Panel A is a vertical section taken across the terminal crest and shows the feature of the sinus node. Panel B is a section cut across the antero-inferior buttress of the atrial septum, showing the features of the atrioventricular node in attitudinally appropriate orientation. Panel C and D are high power images of the right and left sides of the ventricular septum showing the ventricular bundle branches. The bundle branches satisfy all three of the Aschoff-Monckeberg criteria, since the specialized cardiomyocytes are also recognized as being insulated by sheaths of fibrous tissue from the adjacent working cardiomyocytes of the ventricular septum

serving the same purpose for atrial conduction as the insulated pathways subserved for conduction within the ventricular mass.¹⁰ But no histological evidence was provided to support this speculation. The major proponent of this alleged 'specialization', furthermore, paid scant recognition to the insulation of the ventricular pathways as having been chosen to represent the paradigm of a conducting tract.^{3,4} As is shown in *Figure 1*, even at low magnifications, it is possible histologically to recognize the sinus and atrioventricular nodes. Examination of

histological sections prepared in serial fashion then shows that there are no insulated pathways within the atrial walls that provide discrete tracts between the nodes.^{9,11} It is the aggregation of the working cardiomyocytes in parallel fashion, as emphasized by Bachmann,⁸ which provides the substrate for the preferential conduction known to occur through the atrial walls. The sinus impulse is transmitted via areas in which the working atrial cardiomyocytes are aggregated in parallel fashion. These atrial myocardial bundles electrically bridge the nodal tissues through

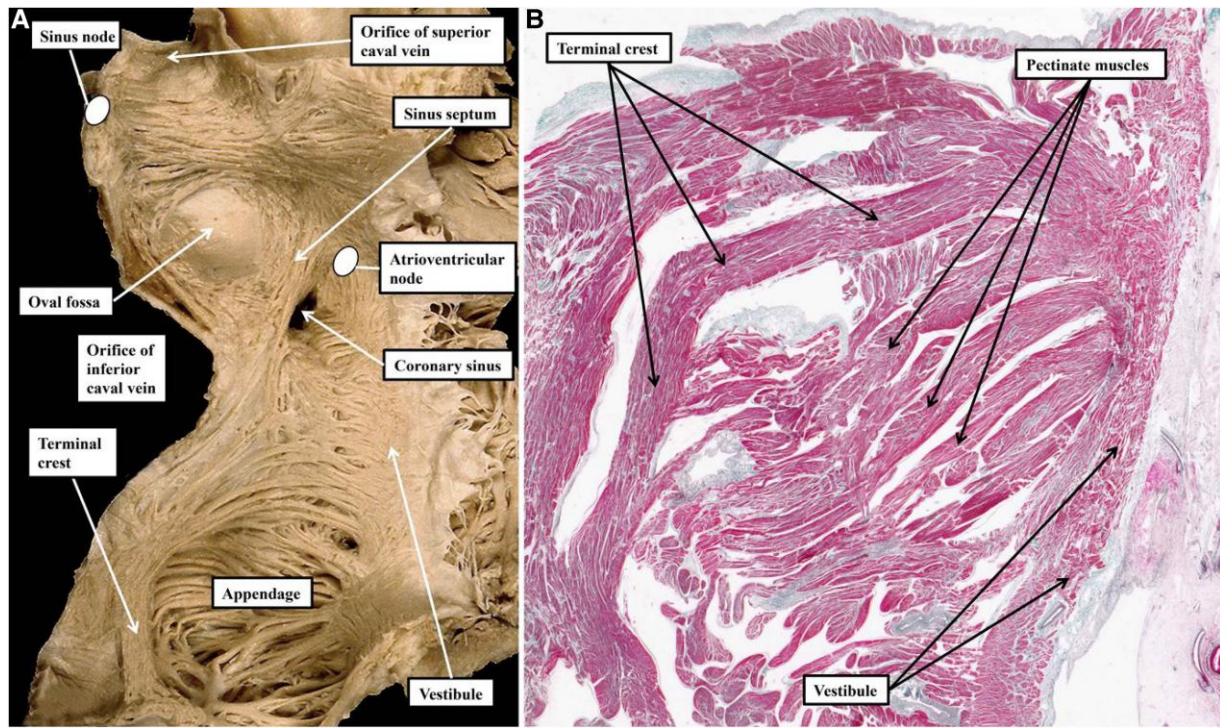


Figure 2 In Panel A, the right atrium of an adult human heart has been opened by making a cut through the posterior wall of the systemic venous sinus and reflecting the parietal component of the chamber inferiorly. The endocardial lining of the walls has then been removed to show the 'grain' of the cardiomyocytes, comparable to the grain seen in wood, produced by the alignment of the aggregated cardiomyocytes making up the walls. Panel B is a histological section, taken to parallel the arrangement as seen in panel A, showing the aggregation of the working cardiomyocytes in the terminal crest, pectinate muscles, and vestibule of the right atrium. The section, cut in the frontal plane, has been reorientated so as to match the arrangement shown in Panel A

intercellular gap junctions that provide the cell-to-cell connections.¹² Diffusion tensor magnetic resonance imaging, a technique that uses water diffusion as a probe to image orientation of the tissues, has demonstrated a set of major areas with regional heterogeneity across the atrial wall with regard to the extent of differences in the measurement of the angulation of the identified components across the wall.¹³ The technique, however, lacks the resolution to identify individual cardiomyocytes but rather tracks the environment in which the cardiomyocytes are housed. It supports the notion of 'preferential highways', matching the evidence of aggregation of the working cardiomyocytes as revealed by classical histology.

Terminal crest

The crest, also known as the 'crista terminalis', marks the border between the pectinated wall of the appendage and the extensive systemic venous component spanning between the caval veins, with the latter part being smooth-walled. The working cardiomyocytes within the terminal crest, which itself represents one of the internodal areas, are similarly aggregated together in parallel fashion, thus favouring preferential conduction (Figure 2).¹⁴ The junctional areas of the terminal crest with the pectinate muscles, and with the intercaval area, in contrast, are characterized by a crossing and nonuniform architecture of the contained

cardiomyocytes.¹⁵ The crest itself has high levels of collagen, expresses low conductance connexins, and displays cellular automaticity.¹⁴ It is an area of marked anisotropy due to poor transverse cell-to-cell coupling.^{16,17} The velocity of propagation within the crest is approximately 10 times greater in the longitudinal than in the transverse direction, this being the essence of preferential conduction.¹⁵ Ultrastructural analysis has also shown that its myocytes are connected by numerous small gap junctions, which are part of relatively simple and straight intercalated disks. Northern blot analysis detected approximately equivalent amounts of mRNAs encoding the gap junction channel proteins connexin 43 and connexin 45, but approximately four times more connexin 40 mRNA than in the left ventricle.¹⁵ In terms of histology, therefore, the crest is made up of working atrial cardiomyocytes, albeit underscoring preferential conduction.

Bachmann's bundle

As with the terminal crest, it is the aggregation of working atrial cardiomyocytes that underscores the primary pathway for spread of conduction to and within the left atrium, emphasizing again the significance of the alignment of the cardiomyocytes in producing preferential conduction.^{8,18–20} The working cardiomyocytes that make up the anterior part of the superior

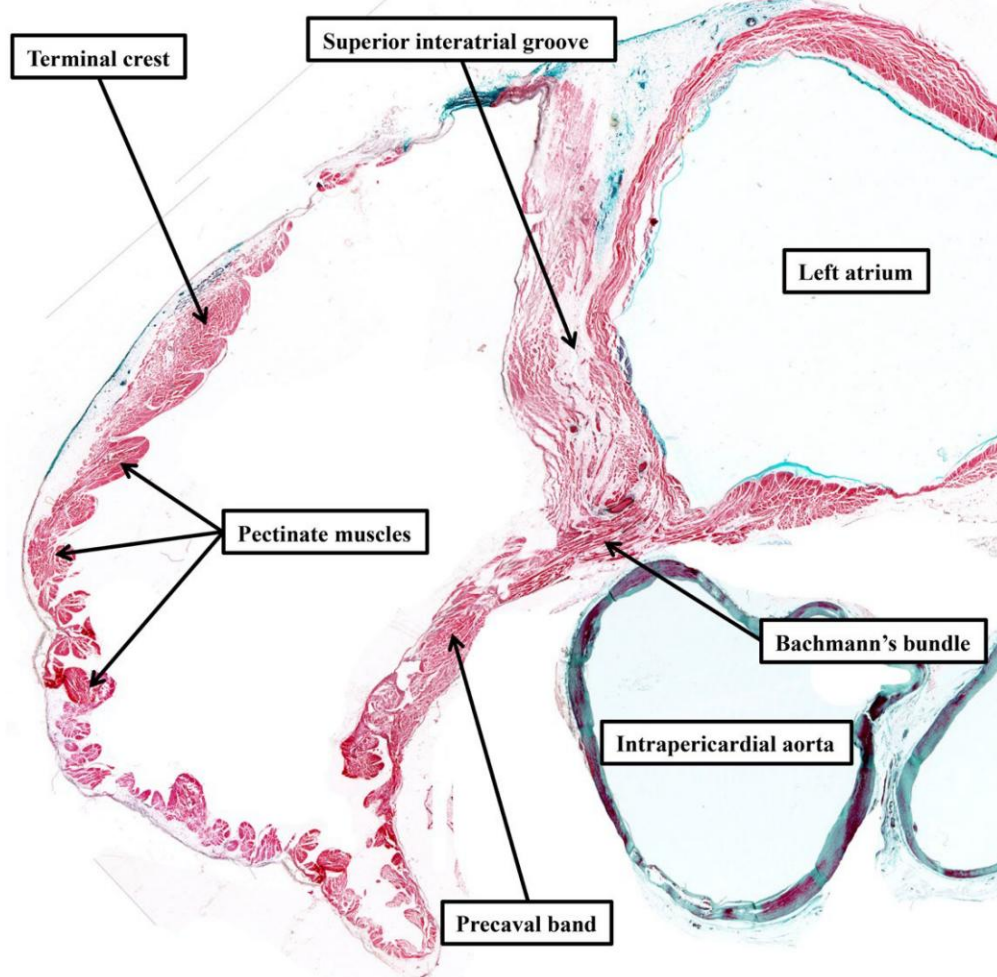


Figure 3 The image shows a histological section taken across the short axis of the atrial chambers, with the anterior walls seen to the bottom of the image and the posterior walls at the top. The right side is viewed to the left hand. The section shows well the parallel alignment of the aggregated working cardiomyocytes in the precaval band, the boundaries of the superior interatrial groove, and Bachmann's bundle. Comparison with [Figure 1](#) shows that none of the cardiomyocytes satisfy the criteria established by Aschoff and Monckeberg for histological 'specialization'

interatrial groove are aggregated together in a parallel fashion, as are those of the precaval band, thus producing the preferential pathway for conduction from the sinus node to the left atrium known as Bachmann's bundle ([Figures 3 and 4](#)).²¹ The cardiomyocytes making up the bundle, when assessed using standard histological sections, are indistinguishable from those making up the remainder of the walls of the left atrium ([Figure 3](#)).²² The cardiomyocytes within the area recognized as the 'bundle' are not themselves insulated from the adjacent working atrial cardiomyocytes. In this regard, the increased interatrial conduction delay that is produced by conventional pacing of the right atrial appendage, as indicated by a wider P wave than observed during sinus rhythm, may be associated with an increased risk of developing atrial fibrillation.²³ Specific pacing of Bachmann's bundle is now increasingly recognized as a promising modality, which theoretically might restore biatrial resynchronization and reduce atrial arrhythmias.^{24,25} Stimulation

close to these preferentially conducting pathways might enhance the efficacy of pacing in the AAI/R mode.

Cavotricuspid isthmus

The pectinated inferior area of the septal aspect of the right atrium is an integral part of the cavotricuspid isthmus. This is yet another area providing preferential conduction due to the parallel arrangement of its contained working cardiomyocytes. The pectinated area makes up the largest middle part of the isthmus, but it also has a smooth anterior myocardial component, which is part of the atrial vestibule, and a posterior fibrous part, where the isthmus abuts the wall of the inferior caval vein ([Figure 5](#)).^{26,27} As was the case with the terminal crest, and Bachmann's bundle, the vestibular myocardium is composed of working rather than specialized cardiomyocytes. High-resolution mapping has indicated

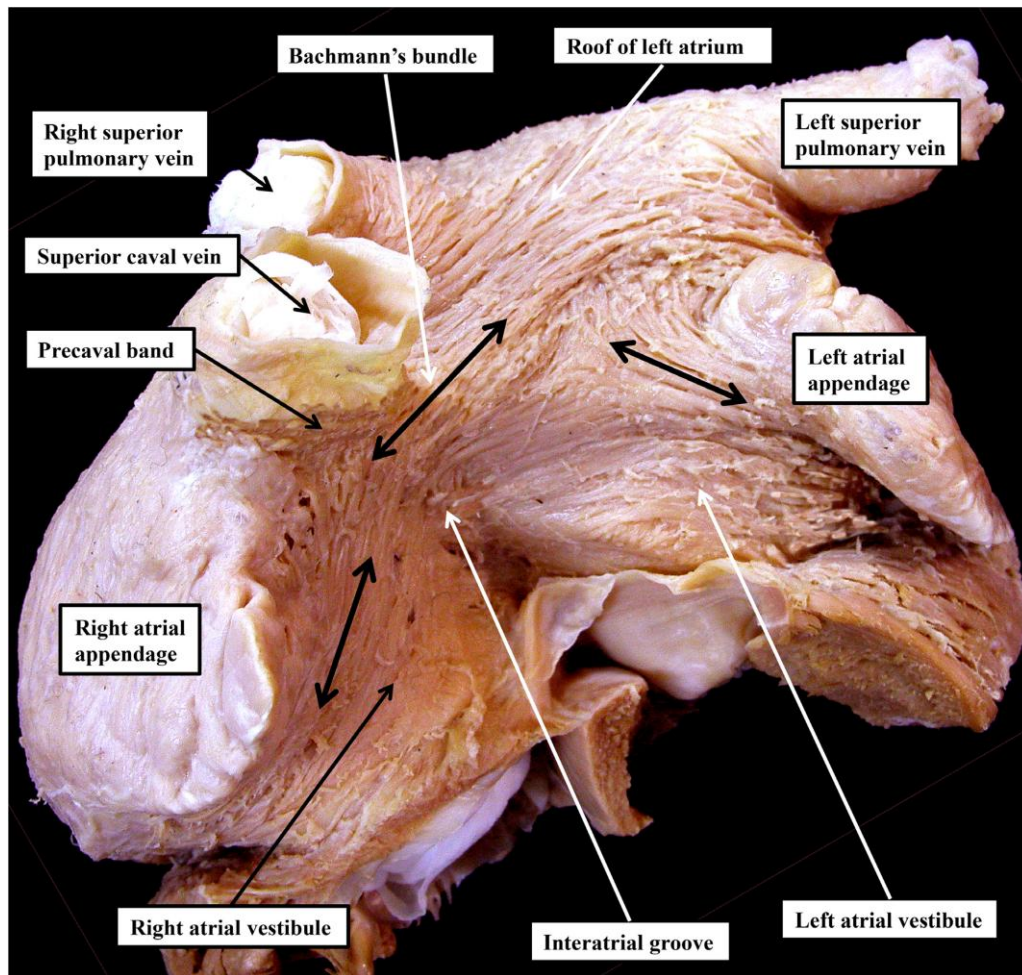


Figure 4 A view of the atrial chambers of an adult heart which has been removed from the remainder of the heart at the level of the atrioventricular junctions. The heart is shown as seen from the front in anatomically appropriate orientation. A dissection has been made by removing the epicardium rather than the endocardial lining to show the 'grain' produced by the aggregation of the cardiomyocytes (double headed black arrows). The precaval band continues as Bachmann's bundle to provide the major pathway for activation of the walls of the left atrium

that conduction velocity at the cavotricuspid isthmus is twice as low as atrial conduction velocities outside the isthmus.²⁸ The nonuniform arrangement of the terminal ramifications of the terminal crest may account for the delay, and for the known discontinuity, in the spread of the excitatory wavefront which uses the isthmus as part of the circuit for atrial flutter. Double or fractionated atrial potentials can be recorded along the isthmus and Eustachian valve and ridge.²⁹

Septal isthmus

The septal isthmus is the area between the orifice of the coronary sinus and the hinge of the septal leaflet of the tricuspid valve (Figure 5). It is of particular electrophysiological importance since, although it again consists of working rather than specialized cardiomyocytes, it facilitates the circuit of atrioventricular nodal re-entrant tachycardia. As such, it is recognized as representing the optimal site for catheter ablation of the arrhythmia.^{22,30}

Atrial inputs to the atrioventricular node

The major atrial inputs to the compact atrioventricular node are found within the vestibules of the tricuspid and mitral valves, which form the right-sided and left-sided atrial boundaries of the pyramid of Koch (Figure 6).³¹ Tracking the inputs through histological sections prepared in serial fashion shows how the inferior extensions merge to form the compact atrioventricular node.³² The compact node itself is recognized primarily because of the obvious organization of its contained specialized cardiomyocytes, which form a half-oval set against the supporting plane of fibrous atrioventricular insulation (Figure 1B). The specialized cardiomyocytes are linked together in the form of a random three-dimensional meshwork, with no obvious alignment of their contained chains. The nodal cardiomyocytes are smaller than the adjacent atrial working cardiomyocytes. Unlike the cardiomyocytes of the ventricular components of the conduction

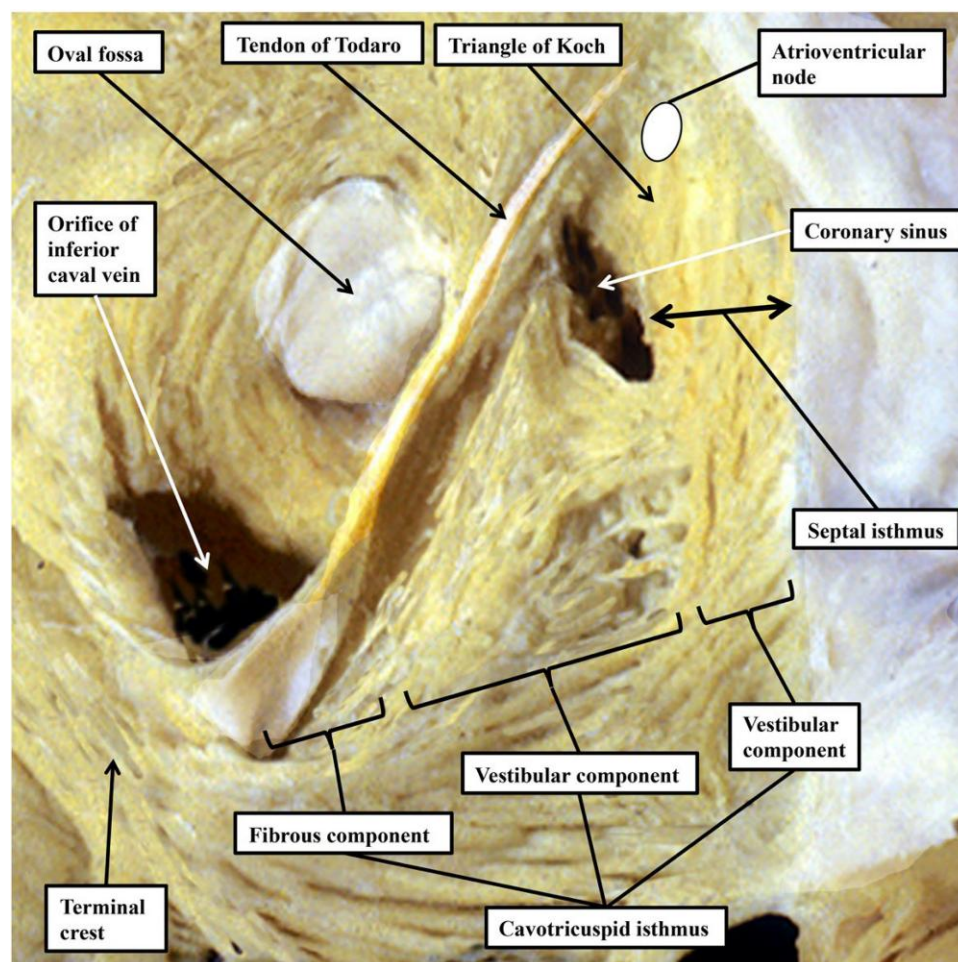


Figure 5 The specimen has been dissected to reveal the 'grain' produced by the aggregation of the working cardiomyocytes making up the atrial walls. In this specimen, the atrial walls are viewed as shown in [Figure 2](#), but with focus specifically on the arrangement of the cavotricuspid and septal isthmuses

axis, however, the nodal cardiomyocytes are not insulated from the inputs provided by the working atrial cardiomyocytes ([Figure 6](#)). The right inferior extension is a combination of atrioventricular canal myocardium and the atrial component of the primary ring. Hence, it retains its 'specialized' histological characteristics. The leftward extension is exclusively atrioventricular canal myocardium, and is not 'specialized' when assessed using standard histological stains.³³ This notion is also supported by connexin genotyping studies.³⁴ A connexin genotyping study in four human hearts has identified the right inferior extension as an area of high connexin 43 (Cx43) expression and, consequently, faster conduction than the node. Expression of Cx43, however, was low in the left extension.³⁵ Thus, although the cardiomyocytes forming these inputs may be histologically discrete, it is again mainly the manner of their aggregation which permits them to be recognized as being 'specialized'. This is because they can be traced from section to section within the serially prepared histological datasets. They are not insulated from the adjacent atrial working cardiomyocytes ([Figure 6A](#)). They are nodal extensions, rather than specific tracts. Additional inputs merge with the node from the buttress of the atrial septum,

which forms the posterior boundary of the pyramid of Koch ([Figure 6B](#)). Atrio-nodal connections are usually provided by the central part of the septum, but they can also take their origin from the superficial or left side of the septum.³⁶ These septal connections, and particularly the final one observed before the node becomes insulated as the bundle of His, are composed of ordinary working myocardium. Considerable histological and electrophysiological evidence now exists to show that the last connection between the working cardiomyocytes and the compact node is part of the substrate of the fast component of the circuit of nodal re-entry tachycardia.^{22,37,38}

Pulmonary veins

The pulmonary veins enter the cavity of the left atrium cranially, joining its roof on either side to form the atrial dome ([Figure 7](#)).^{39–42} The myocardial mural architecture is particularly complex at the venoatrial junctions, producing a three-dimensional intermingling of aggregated individual cardiomyocytes.^{25,41,43} A similar three-dimensional intermingling of the

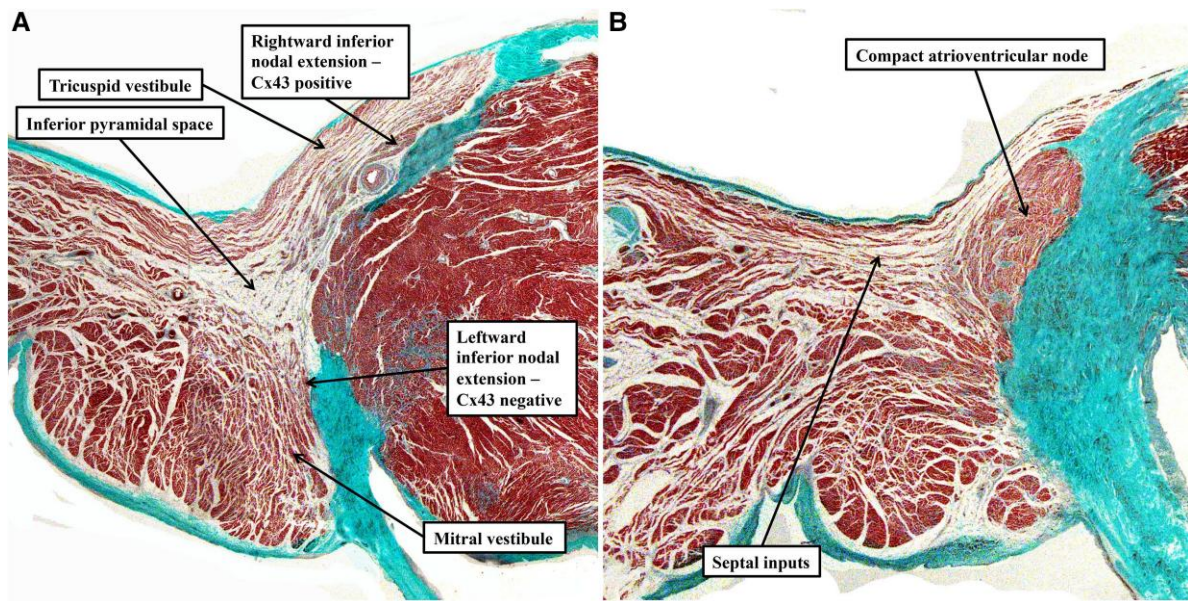


Figure 6 The images, viewed in an anatomically appropriate fashion, with the atrial walls to the left hand, and the ventricular septum to the right hand, are sections taken from a serially sectioned dataset prepared to show the make-up of the atrioventricular node. Panel A shows the major inputs to the node through the inferior extensions making up parts of the mitral and tricuspid vestibules and providing the so-called slow pathways into the node. Panel B shows the arrangement when the inferior nodal extensions have joined to form the compact atrioventricular node, with the section also showing the septal inputs. It is the last septal input, not seen in this section, that represents the 'fast' pathway into the node. The cardiomyocytes of the node fulfil the first of the Aschoff–Monckeberg criteria, in that they are histologically distinct from the adjacent working cardiomyocytes. When traced through the serial sections, the node can be followed to reveal its overall arrangement, thus fulfilling the second criterion. The nodal cardiomyocytes, however, are not insulated from the atrial working cardiomyocytes

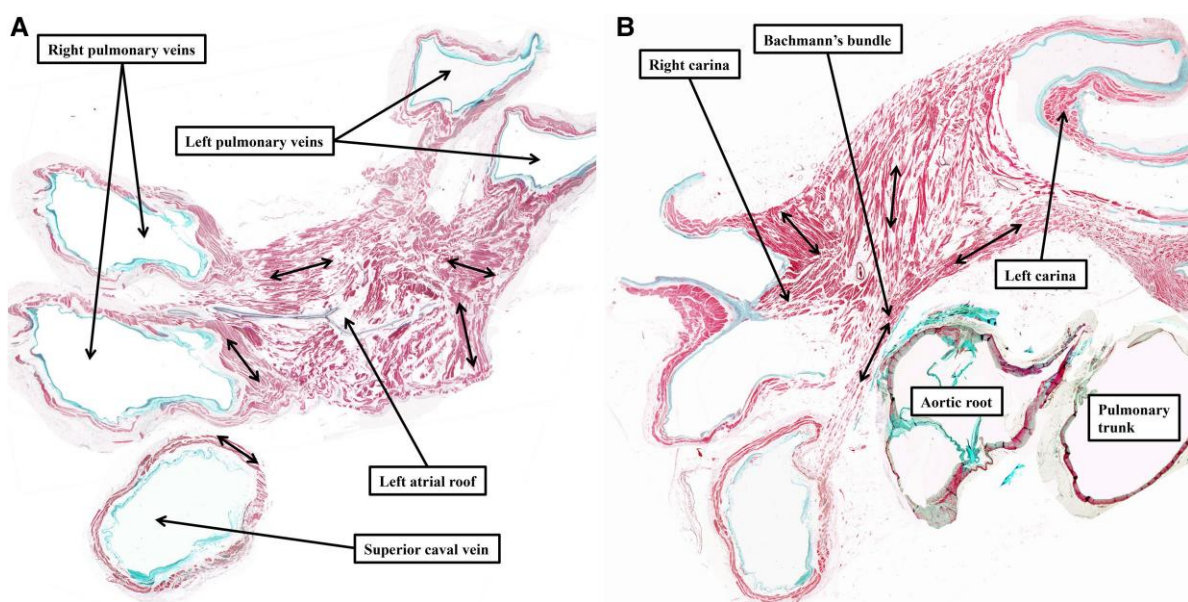


Figure 7 These serial histological sections, orientated as for Figure 3, and stained using the trichrome technique, are taken in the short axis of the atrial chambers. Panel A shows the three-dimensional arrangement of the aggregated cardiomyocytes making up the superficial layer of the roof of the left atrium and the venoatrial junctions. Panel B is a deeper section. It shows the continuity through Bachmann's bundle with the dome of the left atrium. The double-headed arrows show the parallel alignment of the cardiomyocytes that underscores preferential conduction. None of the cardiomyocytes, however, satisfy the criteria of Aschoff and Monckeberg for histological specialization

aggregated cardiomyocytes is even more obvious in the roof of the left atrium (Figure 7). Thus, there is an obvious 'grain' in the fashion of aggregation of the working atrial cardiomyocytes that make up the mural myocardial components. In this regard, the term 'grain' is used to parallel the arrangement as seen in sections of timber or furniture, again reflecting the pattern of the tissues making up the wood. The parallel arrangement is presumed to favour preferential conduction along the long axis of the grain. The primary activation of the left atrium is through Bachmann's bundle, which as already established is made up of cardiomyocytes aggregated together in parallel fashion. Some have identified the major pathway across the atrial dome as the 'septopulmonary bundle'.^{44,45} This is deceptive. The aggregated cardiomyocytes forming the superficial layer of the atrial wall are part of a much more complicated three-dimensional array of cardiomyocytes that form the overall thickness. This is well seen in the histological sections, which show marked changes in the observed 'grain' (Figure 7). As emphasized, the manner of aggregation of the cardiomyocytes within the sleeves is similarly complex, with obvious changes in 'grain'. Within the thickness of the walls, some of the cardiomyocytes are aggregated in circumferential fashion, but these are interconnected with other aggregates having a longitudinal orientation.⁴² Such an arrangement may theoretically lead to anisotropic conduction between the bundles, which in itself can act as a focal trigger or a substrate for micro re-entry.

Interatrial connections

Although Bachmann's bundle is the dominant interatrial pathway,¹⁸ it is not the only area in which working cardiomyocytes provide bridges between the walls of the right and left atrial chambers. The cardiomyocytes making up the infolded rims of the fossa provided multiple interatrial pathways.⁴⁶ Those in the right atrial sides of the posterior-inferior and superior portions, furthermore, can be traced towards the atrioventricular node.⁴⁷ Additional interatrial muscular connections can be found between the right pulmonary veins and the walls of the right atrium and inferiorly between the walls of the coronary sinus and the inferior wall of the left atrium.^{46,47}

Cardiomyocytes of the ventricular septum

Apart from the ventricular components of the atrioventricular conduction axis, the Aschoff-Monckeberg criteria are just as important in establishing the nature of the remaining cardiomyocytes that make up the ventricular septum. Unlike the ventricular walls, which have often been presumed to form by 'compaction' of pre-existing trabecular myocardium, the muscular ventricular septum is formed by coalescence of the initial embryonic ventricular trabeculations.⁴⁸ The cardiomyocytes forming the branches of the atrioventricular bundle are part of this coalescing structure. The bundle branches, however, are recognized as being 'specialized' because they are insulated by fibrous sheaths from the underlying working cardiomyocytes that make up the bulk of the muscular ventricular septum (Figure 8A). There is nothing 'special' about the septal cardiomyocytes as

opposed to the bundle branches when they are assessed using the criteria proposed by Aschoff and Monckeberg.^{1,2} Recent evidence has shown, nonetheless, that along with the cardiomyocytes making up the bundle branches, some of the cardiomyocytes within the septum can be identified when sections are stained to show the myosin light chain 4 (MYL4) gene that is identified as a Purkinje fibre-specific cell marker.⁴⁹ This finding has then been extrapolated to suggest that such cardiomyocytes are part of the ramifications of the left bundle branch, with the cardiomyocytes themselves being described as 'Purkinje cells'.⁵⁰ No evidence has been provided, however, to show that the alleged 'Purkinje cells' are in continuity with the insulated cardiomyocytes that make up the bundle branches. Nor are the cardiomyocytes themselves appropriately described as being 'Purkinje cells'. The cells described by Purkinje are found only in ungulate hearts. The cardiomyocytes within the insulated bundle branches are no larger than the adjacent working cardiomyocytes. Thus, it is rather premature to suggest that cells sharing the content of the MYL4 gene might play a role in explaining the success of left bundle branch area pacing. Nor is there any evidence to show that the marked cardiomyocytes are joined together so as to favour preferential conduction. This is in contrast to the cardiomyocytes forming the fasciculo-ventricular connections, also known as superior septal pathways.⁵¹ These cardiomyocytes do satisfy the criteria of Aschoff and Monckeberg as being anatomically specialized and are also aligned to favour preferential conduction to the ventricular septal crest (Figure 8B).⁵² The finding that cardiomyocytes within the muscular ventricular septum share a gene with insulated cardiomyocytes of the bundle branches, nonetheless, is of interest, albeit that the presence of the gene, in itself, has not thus far been shown to convey any 'special' function. It is just as likely that the presence of fasciculo-ventricular connections, also known as a superior septal pathways, could explain the efficacy of so-called left bundle branch area pacing.

Conclusions

There are areas in the heart, and especially the atrial walls, where working cardiomyocytes are aggregated together in a parallel fashion, producing pathways favouring preferential conduction. The pathways, however, lack the insulation which is the feature defining the 'specialized' nature of the ventricular components of the cardiac conduction system. In terms of histology, apart from those making up the nodes, the atrial cardiomyocytes are 'ordinary'. In many areas, nonetheless, they are obviously aggregated together in parallel fashion. This feature, together with particular characteristic of the cell-to-cell connections, creates the conditions for functional as opposed to histological specialization. In other words, the ability for the majority of atrial cardiomyocytes to conduct preferentially is not a consequence of their histological appearance. In this context, the term 'preferential' as opposed to 'specialized' conduction is pertinent (Graphical Abstract). New techniques, such as laser histological capture which permits recognition of such cells in histological sections, transcriptomics which allow the assessment of the genetic make-up of individual cells, and the integration of single cell RNA-sequence and spatial transcriptomics that provides an

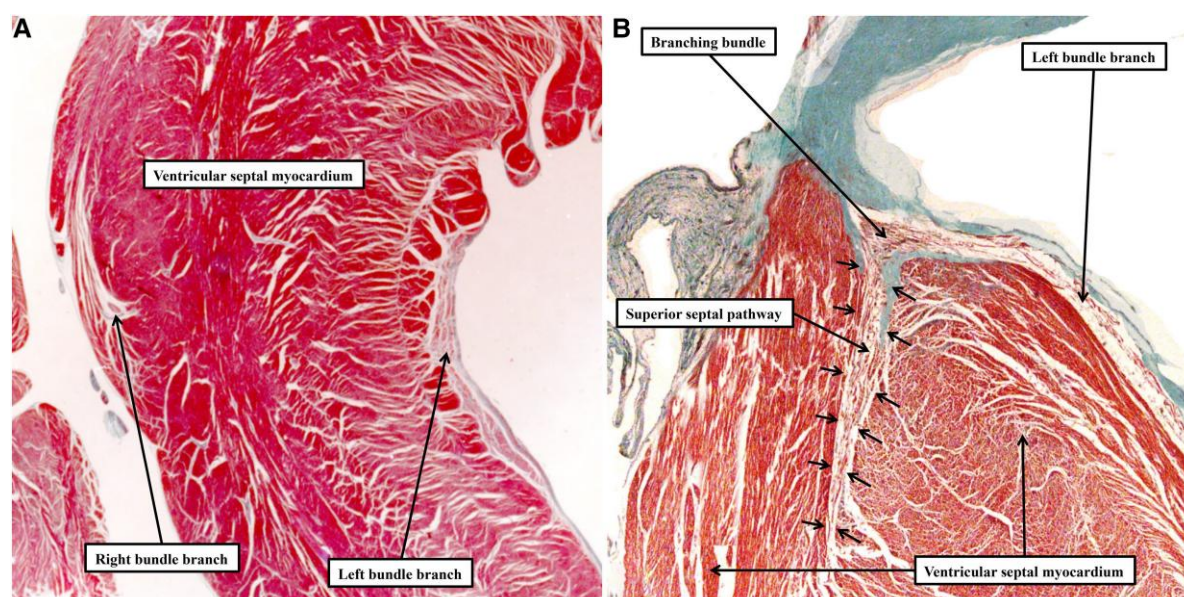


Figure 8 The histological sections show how the Aschoff–Monckeberg criteria can also be used to identify the ‘specialized’ cardiomyocytes within the ventricular muscular septum. Panel A, taken in the short axis of the ventricular mass, with the right side seen to the left hand, shows a lower power image of the right and left bundle branches, shown at higher magnifications in [Figures 1C and D](#). It is only the bundle branches that satisfy the criteria for histological specialization. Panel B shows a section through the crest of the muscular septum in ‘four chamber’ orientation. It shows how a fasciculo-ventricular connection, also known as a superior septal pathway (between single-headed arrows), extends directly as a histologically specialized bundle into the septal crest. Its full extent can be tracked through the serial histological sections, thus fulfilling the second criterion of Aschoff and Monckeberg. The presence of such superior septal pathways provides a potential explanation for the efficacy of the so-called left bundle branch area pacing

overview of cardiac tissue organization and intercellular communication, are being developed and implemented.⁵³ With the advent of such techniques, it may well prove possible to identify particular features of histologically indistinct cardiomyocytes that influence conduction and hence perhaps underscore a mode of ‘functional’ specialization. Until, and when, such new evidence is provided, it is preferable to account for conduction across the atrial walls as being preferential. This is in marked contrast to the ‘specialized’ nature of ventricular conduction, known to be the consequence of the insulation of the components of the conducting tracts from the adjacent working ventricular cardiomyocytes.

Declarations

Disclosure of Interest

Nothing to declare.

Data Availability

No data were generated or analysed for or in support of this paper.

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Nothing to declare.

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